PALATE

AN EMOTION-AWARE, ENTROPY-REDUCING FOOD RECOMMENDATION ENGINE FOR LATENT HUMAN INTENT

TECHNICAL AND PRODUCT VISION

CRAVE

ADAPTIVE SEARCH & BAYESIAN INFORMATION THEORY

The Paradigm Shift: *Solving Latent Intent*

- **Traditional Food Apps Solve the Wrong Problem:** Recipe sites optimize for SEO, diet apps for metrics, meal planners for schedules.
- **The Real Human Question:** Users rarely open an app knowing what they want. They ask: *"What do I feel like eating?"*
- **The Architectural Difference:** Palate is not a searchable database. It is an **adaptive search engine over human food preference**.
- **Core IP:** The product is not the recipe collection—it is the recommendation algorithm.



The Akinator Model: *Interactive Discovery*

Active Elimination

Instead of manual scrolling or static keyword filters, the engine asks a sequence of high-gain questions. Each answer dynamically cuts down the search space.

Information Theory

Questions are chosen dynamically based on which query promises the largest expected reduction in candidate uncertainty (maximum entropy reduction).

Zero Manual Search

The user never searches a database manually. The system actively discovers the target dish *together* with the user in minimal steps.

Search Space Compression: *100,000 to 5*

How the engine dynamically navigates down a decision tree through intent reduction:

100k
INITIAL SET
All Recipes

›

18k
"SOMETHING WARM"
-82% Space

›

5.4k
"COMFORT FOOD"
-70% Space

›

900
"UNDER 45 MINS"
-83% Space

›

5
TARGET SET
Pho / Laksa / Ramen

Engine Goal: Continuously select attributes that bisect the candidate set efficiently, stopping as soon as recommendation confidence crosses a threshold.

Mathematical Core: *Bayesian Updating*

$$P(\text{recipe} \mid \text{user responses})$$

Posterior probability updated continuously over interaction sequence

Instead of rigid, binary filters that permanently eliminate recipes early, Palate maintains a continuous probability distribution across the candidate set.

- **Maximal Entropy Reduction:** Prompts prioritize attributes that split the candidate distribution effectively rather than low-information questions (e.g. "Do you like garlic?").
- **Soft Probability Shifts:** Selecting "Comfort Food" boosts probability vectors for soups & stews while gradually dampening light salads.
- **Uncertainty Representation:** Gracefully handles ambiguous user feedback without hard failure states.

Mood Driven Search Trajectory

Emotion Overrides Habit: Traditional engines over-index on historical behavior. But a stressed user who usually eats salad will crave ramen, while an energized user wants fresh fruit.

Primary Signal: Mood determines the directional vector of the search tree itself before secondary attributes (time, budget) are calculated.

> **Key Architectural Rule:** Historical preferences influence recommendations, but current emotional context governs them.



System Architecture: *Modular Pipeline*

1

User Context

Session initiation & emotion query

2

Question Generator

Entropy-based candidate prompts

3

Bayesian Update

Probability distribution recalculation

4

Ranking & Explain

Multi-model scoring & natural rationale

Loose Coupling

Each pipeline stage operates as an independent micro-module, allowing seamless iteration from heuristics to ML models.

Configurable Scoring

Weights for mood, context, history, and discovery bonuses can be tuned independently in real-time.

User Representation: *Three State Models*

Long-Term Model

Evolves Slowly: Captures baseline attributes like spice tolerance, skill level, dietary restrictions, and ingredient dislikes.

Session Model

Ephemeral Context: Captures state at execution time—mood, weather, available prep time, servings, and fridge contents.

Behavior Model

Continuous Signals: Implicit evidence derived from recipes viewed, skipped, cooked, substituted, or saved over time.

Recipe Representation: *Multi-Sensory Taxonomy*

Dimension	Key Metadata Vectors	Recommendation Purpose
Structural	Cuisine type, prep duration, difficulty, ingredient graph	Filters physical constraints and baseline feasibility
Sensory	Warmth, umami, richness, crunchiness, spiciness, creaminess	Matches physical taste cravings & texture desires
Emotional	Comfort, indulgent, celebratory, adventurous, nostalgic	Aligns directly with user emotional state
Contextual	Rainy day, late night, quick lunch, workout recovery	Surfaces meals fitting environmental conditions

Knowledge Space: *Graph Navigation*

Cuisine Proximity Graph

Cuisines are connected through shared flavor profiles rather than static rigid categories:

```
Japanese → Ramen → Pho → Laksa → Paya → Stews
```

Enables smooth neighborhood exploration across international boundaries.

Ingredient Substitution Graph

Dynamic ingredient substitution generated via graph traversal:

```
Butter → Ghee → Olive Oil
Miso → Doenjang → Soy Sauce
```

Adapts recommendations seamlessly based on pantry availability.

Scoring Engine & *Serendipity Bonus*

Scoring Formula Components

```
Score = Mood_Match + Context_Match + Behavior_Sim +
History + Discovery_Bonus - Constraint_Penalties
```

- **Preventing Feedback Loops:** Naive algorithms repeat identical choices. Palate explicitly optimizes for expected delight rather than pure frequency.
- **Discovery via Proximity:** A ramen lover is introduced to Laksa or Soto Ayam through flavor graph proximity, not pure randomness.

Roadmap & *Native Explainability*

💬 Built-in Explainability

"We recommended Laksa because today you wanted something comforting, warm, and moderately adventurous, and you enjoy noodle soups."

Builds trust with users and provides transparent debugging telemetry for developers.

🚀 Future ML Evolution

The architecture assumes handcrafted heuristics will evolve into learned components:

- Graph Neural Networks (GNNs) over ingredients
- Multimodal food embeddings
- Contextual Bandit & Reinforcement Learning

Key Architectural *Takeaway*

> "The MVP already proves the core innovation: an entropy-optimizing Bayesian engine. Future iterations refine the data quality and machine learning, not the foundational algorithm."

Proven IP Core
Algorithms remain constant; continuous learning enhances recommendation fidelity over time.



Goal-Based Eating & Macro Alignment

The Standard Fitness App Problem

Most fitness and nutrition apps calculate exact daily requirements but leave you stranded when deciding what to actually prepare or order.

CALORIES **2,400 kcal**	PROTEIN **180g**
CARBS **250g**	FAT **70g**

Result: "How much should I eat?" is answered. "What should I eat?" remains unsolved.

The Palate Paradigm Shift

From "How much should I eat?" → "What should I eat?"

You specify your fitness targets, macro splits, and time constraints. Palate generates exact, chef-grade dish recommendations optimized for your exact nutritional targets without sacrificing flavor.

- ✓ Macro Perfect Matches
- ✓ Prep Time Filters
- ✓ Zero Waste Prep

Craving Search & Natural Language

USE CASE 02

"I Know What I Want. I Just Can't Find It."

The Friction: Traditional search filters rely on rigid keyword tags (e.g. "Asian", "Spicy", "Noodle"). When you search for nuanced sensory desires:

> *"Something rich, spicy, noodle-based, with a coconut broth... but not too heavy."*

Legacy platforms return **0 results** or irrelevant listings.

Palate's Vector Solution

Multimodal NLP vector embeddings map subjective sensory queries directly to dish attribute vectors—instantly unlocking dishes like *Spicy Seafood Laksa* or *Khao Soi*.



Active Discovery & Entropy Reduction



USE CASE 03

"I Want Something. I Just Don't Know What."

Overcoming Decision Fatigue & Monotony

Users fall into 5-dish repeat cycles because evaluating hundreds of recipes creates cognitive overload. Browsing static menus amplifies fatigue.

Conversational Preference Elicitation

Palate initiates 6–10 dynamic adaptive queries (e.g., texture, heat tolerance, prep effort) to drastically narrow candidate probability space and deliver surprising, delightful matches.

One Platform. Three Pillars Solved.

Goal-Based

Translates rigid target macros and calorie budgets into satisfying, real-world dishes.

MACRO PRECISION

Craving Search

Parses complex natural language sensory requests into deep attribute vector matches.

SEMANTIC ALIGNMENT

Active Discovery

Calculates entropy-minimizing prompts to guide indecisive users to optimal new culinary experiences.

ENTROPY REDUCTION

UNIFIED AI CORE

Powered by Bayesian Inference & Dynamic Vector Personalization

STRATEGIC EXECUTION

MVP Scope & *Product Evolution*

Transitioning from an entropy-reducing research prototype into an autonomous, multimodal personalization engine.

Phase 1: Research Prototype (~250 Recipes) → Phase 2: Autonomous Scaling (100k+ Recipes)

MVP (Research Prototype): *Core Capabilities*

Bayesian Inference Engine

Maintains a dynamic probability distribution over all candidate recipes throughout user interaction.

Entropy-Driven Queries

Evaluates 500+ candidate questions to select the exact prompt yielding maximum information gain.

Rapid Convergence

Reaches accurate target dish recommendations in just 6–10 adaptive questions without manual browsing.

Semantic Attribute Vector

Handcrafted dataset of ~250 recipes enriched across structural, sensory, and emotional dimensions.

Explainable Rationale

Provides transparent confidence scores and natural language justifications for every recommendation.

Modular Architecture

Loose coupling ensures each heuristic micro-component can seamlessly swap for learned ML models.

Evolutionary Vector: *Prototype to Scale*

CURRENT MVP

VALIDATION PHASE

Research Engine

- **Dataset Scale:** ~250 Curated Recipes
- **Feature Engineering:** Handcrafted Attributes
- **Question Logic:** Heuristic Information Gain
- **Context Capture:** Explicit Active Prompts
- **Goal:** Validate Core Decision IP

FUTURE PRODUCT

AUTONOMOUS SCALE

Autonomous Engine

- **Dataset Scale:** 100,000+ Unlimited Recipes
- **Feature Engineering:** Multimodal AI Metadata
- **Question Logic:** RL & Contextual Bandits
- **Context Capture:** Implicit Context Sensing
- **Goal:** Scale & Continuous Learning

Planned Evolution: *Scaling & Personalization*

PILLAR 01

Automated Scale

- Expand catalog from **250 → 100,000+** recipes effortlessly.
- Automate metadata extraction using **multimodal AI vision & NLP**.
- Build **GNN similarity graphs** over ingredients and cuisines for smart substitutions.

PILLAR 02

Deep Taste Modeling

- Learn **latent taste representations** beyond rigid human vector tagging.
- Construct **evolving user taste profiles** from long-term behavior.
- Apply **Reinforcement Learning & Bandits** for long-term user satisfaction.

PILLAR 03

Implicit Sensing

- Infer user mood, weather, and time **automatically** without questionnaires.
- Detect pantry ingredients and social setting contexts passively.
- Deliver zero-friction, anticipatory food recommendations.

Questions & Discussion

Transforming food discovery through goal-aligned, active personalization.

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